

SHUBHAM KUMAR

☎ +91-9871805948 ✉ shubh79.nolan@gmail.com 🐙 GitHub 🔗 LinkedIn 🌐 Codolio 🌐 LeetCode

EDUCATION

Indian Institute of Information Technology, Ranchi

Bachelor of Technology in Computer Science and Engineering, *CGPA: 8.15/10*

Ranchi, Jharkhand

2023 – 2027

EXPERIENCE

Practitionerist

September 2025 - May 2026

Full Stack Developer Intern

Remote

- Architected a **multi-service booking engine** (consultations, subscriptions, webinars, classes) handling session allocation, availability windows, and **conflict-aware rescheduling** across overlapping schedules.
- Designed a **multi-layer booking constraint system** — weekly/monthly frequency caps, plan duration, session validity, and subscription quotas — enforced **server-side** to prevent overbooking and quota bypass.
- Built **role-based REST APIs** across 4 dashboards (Admin, Staff, Consultant, Consulatee) with permission control at both **route and resource level**, cutting booking failure rates by **30%** through systematic edge-case handling.
- Integrated **payment gateway flows** and **server-side caching** on availability lookups, reducing repeated database reads on the hottest endpoint and improving API response times by **15%**.

Ziovy

May 2025 - July 2025

Full Stack Developer Intern

Remote

- Built an **admin platform** for rewards, engagement, and campaign management, integrating **GraphQL APIs** for real-time metrics and runtime-configurable campaign rules across 2 production releases.
- Reduced dashboard data-fetch latency by 35% by consolidating multiple GraphQL requests and requesting only the fields each view needed.

SKILLS

Programming Languages: C++, Python, TypeScript, JavaScript, SQL

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

Backend: Node.js, Express.js, NestJS, FastAPI, REST, GraphQL, Socket.io, BullMQ, Prisma, Drizzle ORM

Databases: PostgreSQL, MySQL, MongoDB, Redis

Tools/Platforms: Docker, Nginx, Git, GitHub, Postman, Google Cloud Platform

Frontend: Next.js, React.js, Redux, TanStack Query

PROJECTS

QueueFlow: 🐙 [Source Code](#)

Node.js, TypeScript, Redis, PostgreSQL, Docker

- Built a **distributed job queue on raw Redis primitives** (no BullMQ) — HTTP producer, worker pool and scheduler over **PostgreSQL as system of record**; one Docker image runs any role.
- Guaranteed **at-least-once execution across worker crashes** via atomic BLMOVE handoff, **TTL heartbeats** as the failure detector and a reaper that requeues stranded jobs; retries use **exponential backoff with jitter** into a dead-letter queue.
- Made duplicate execution harmless with **database-backed leases** and **idempotency keys**; **chaos tests** SIGKILL workers mid-job — **320 jobs, zero lost, zero duplicated**, proven by side-effect rows; benchmarked at **~1,000 jobs/s**, bounded by **Postgres commit rate**.
- Added **graceful shutdown** so deploys drain in-flight jobs instead of routing them through crash recovery, plus a live React dashboard with **keyset pagination**, DLQ replay and operator auth.

DevArcade: [Live Link](#) | 🐙 [Source Code](#)

Next.js, TypeScript, PostgreSQL, Clerk, Judge0

- Built a **coding-interview preparation platform** with structured learning paths, chapter-wise progress tracking, and subscription-gated content over a **normalized PostgreSQL schema**.
- Integrated **Judge0** for multi-language code execution (C++, Java, Python), handling **asynchronous submission tokens, result polling, and per-submission resource limits**.
- Implemented **Clerk authentication** with subscription-tier authorization enforced **server-side**, preventing direct-API bypass of premium content, plus an XP-based progression system.

ACHIEVEMENTS

- Earned **Knight** badge on [LeetCode](#) — Contest Rating **1871**, **Top 5.6%** globally; 3-Star on [CodeChef](#) (Max: 1611).
- Solved **800+** **algorithmic problems** across LeetCode, Codeforces, CodeChef, and HackerRank.
- Member of **House of Geeks**, the coding club at IIIT Ranchi.